

# Report

Title: Recovering Access to an EC2 Instance with a Lost Key Pair

## Setup

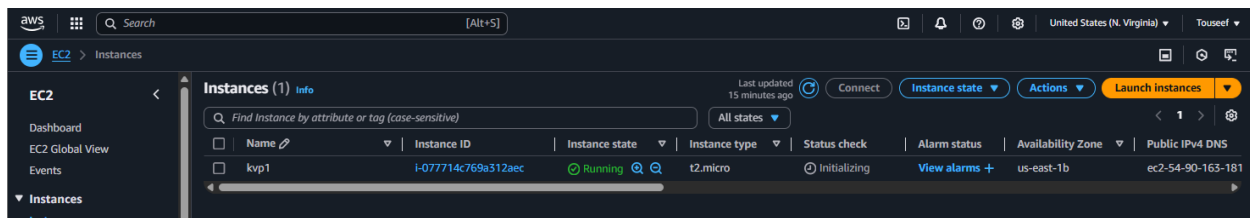
An EC2 instance who is no longer accessible due to lost key pair

Creating a fresh temporary EC2 instance with following configs;

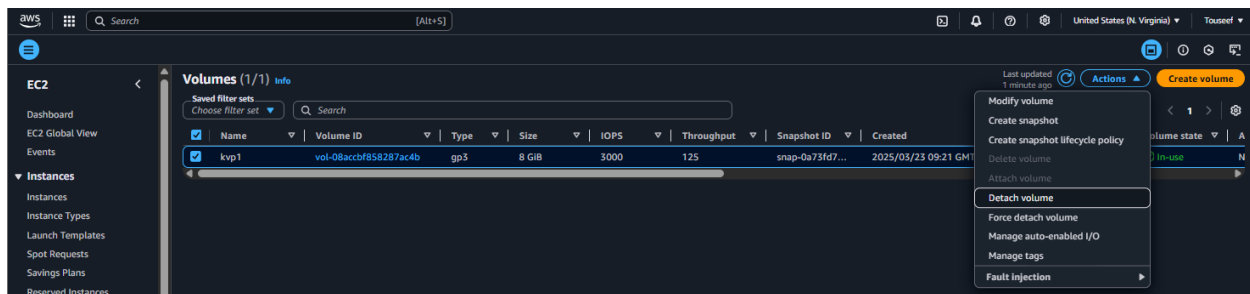
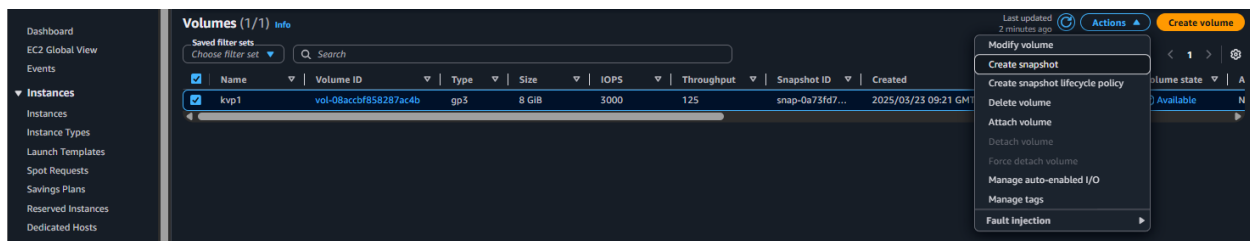
- AMI => Amazon Linux OS
- Instance Type => T3.micro
- Create New Key Value Pair for SSH access (it will download the .pem file on your machine)
- Keep other at Default

## Procedure

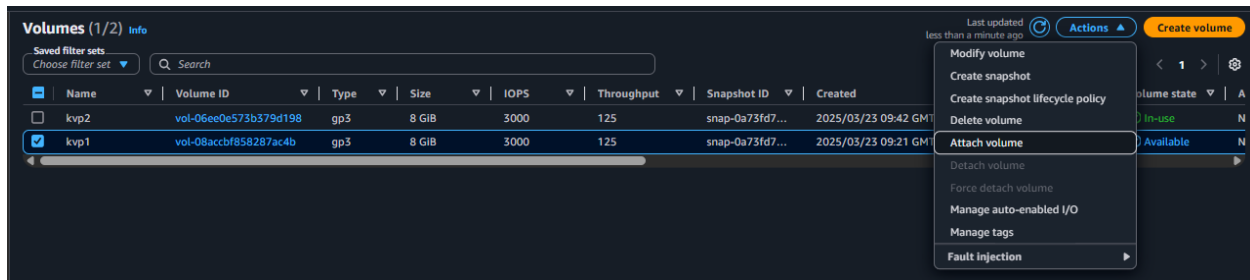
This is the EC2 instance who is no longer accessible.



First step is to stop this instance, create a snapshot of the EBS volume (for backup, in case something goes wrong) and detach its default EBS volume as show below;



Launch a temporary fresh EC2 instance to help restore access to inaccessible EC2 instance. Attach the previous EC2 instance's EBS volume to this new temporary instance as follow;



**Volumes (1/2)** Info

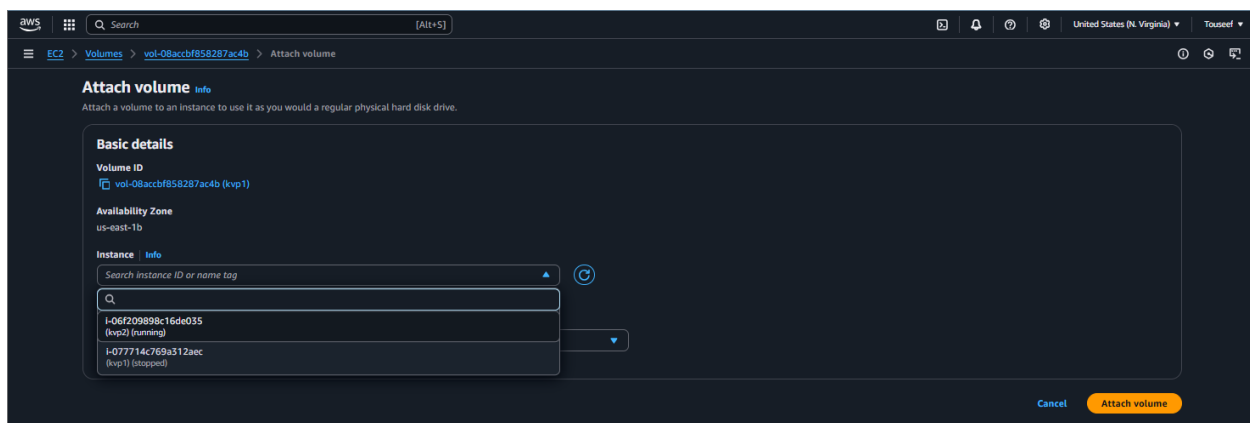
Save filter sets  
Choose filter set

Search

| Name | Volume ID             | Type | Size  | IOPS | Throughput | Snapshot ID     | Created              | Volume state | Availability Zone |
|------|-----------------------|------|-------|------|------------|-----------------|----------------------|--------------|-------------------|
| kvp2 | vol-06ee0e573b379d198 | gp3  | 8 GiB | 3000 | 125        | snap-0a73fd7... | 2025/03/23 09:42 GMT | In-use       | N                 |
| kvp1 | vol-08accbf858287ac4b | gp3  | 8 GiB | 3000 | 125        | snap-0a73fd7... | 2025/03/23 09:21 GMT | Available    | N                 |

Actions

- Modify volume
- Create snapshot
- Create snapshot lifecycle policy
- Delete volume
- Attach volume**
- Detach volume
- Force detach volume
- Manage auto-enabled I/O
- Manage tags
- Fault injection



**Attach volume** info

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

**Basic details**

Volume ID  
vol-08accbf858287ac4b (kvp1)

Availability Zone  
us-east-1b

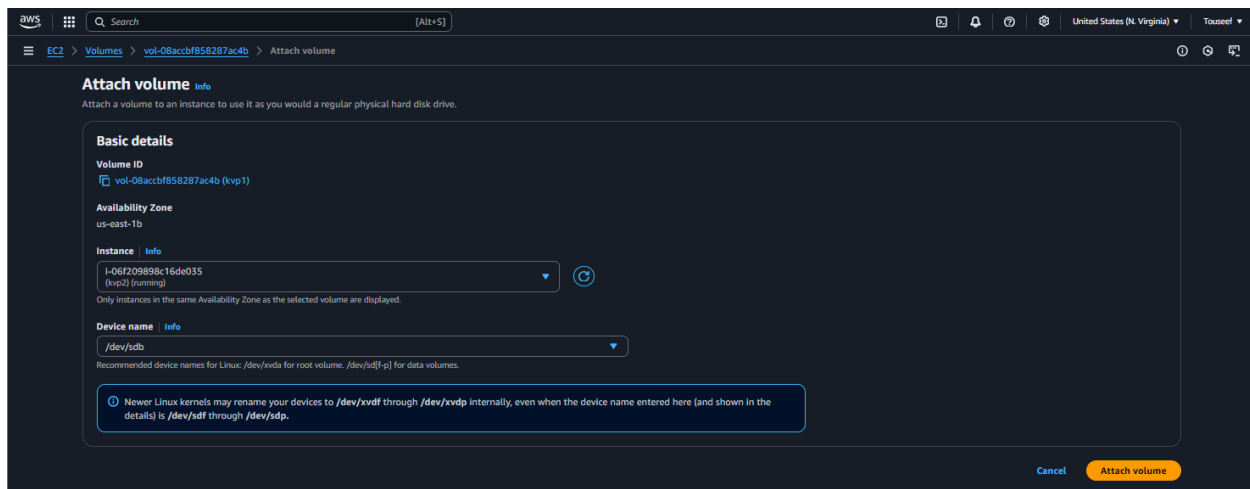
Instance | Info

Search instance ID or name tag

Q

- i-06f209898c16de035 (kvp2) (running)
- i-077714c769a312aec (kvp1) (stopped)

Cancel Attach volume



**Attach volume** info

Attach a volume to an instance to use it as you would a regular physical hard disk drive.

**Basic details**

Volume ID  
vol-08accbf858287ac4b (kvp1)

Availability Zone  
us-east-1b

Instance | Info

i-06f209898c16de035 (kvp2) (running)

Only instances in the same Availability Zone as the selected volume are displayed.

Device name | Info

/dev/sdb

Recommended device names for Linux: /dev/xvda for root volume, /dev/sd(f-p) for data volumes.

ⓘ Newer Linux kernels may rename your devices to /dev/xvdf through /dev/xvdp internally, even when the device name entered here (and shown in the details) is /dev/sdf through /dev/sdp.

Cancel Attach volume

Now SSH to new temporary EC2 instance and mount the newly attached volume as follows;

List block devices, and find the newly attached volume. In my case it is xvdb;

```
[ec2-user@ip-172-31-30-22 ~]$  
[ec2-user@ip-172-31-30-22 ~]$  
[ec2-user@ip-172-31-30-22 ~]$ lsblk  
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS  
xvda         202:0    0   8G  0 disk  
├─xvda1      202:1    0   8G  0 part /  
├─xvda127    259:0    0    1M  0 part  
└─xvda128    259:1    0   10M  0 part /boot/efi  
xvdb         202:16   0   8G  0 disk  
├─xvdb1      202:17   0   8G  0 part  
├─xvdb127    259:2    0    1M  0 part  
└─xvdb128    259:3    0   10M  0 part  
[ec2-user@ip-172-31-30-22 ~]$
```

Switch to root user, and create a new mount location. Also find out the filesystem type of the new volume. The expected file system should be “/dev/xvdb1: Linux rev 1.0 ext4 filesystem data ...” but in our case, the file system is XFS file system. So mention the filesystem in mount command below.

Note:

/dev/xvdb = the whole disk

/dev/xvdb1 = first partition (the actual filesystem is here)

```
[root@ip-172-31-30-22 ec2-user]#  
[root@ip-172-31-30-22 ec2-user]# mount /dev/xvdb1 /mnt/old_volume  
mount: /mnt/old_volume: wrong fs type, bad option, bad superblock on /dev/xvdb1, missing codepage or helper program, or other error.  
[root@ip-172-31-30-22 ec2-user]#  
[root@ip-172-31-30-22 ec2-user]# file -s /dev/xvdb1  
/dev/xvdb1: SGI XFS filesystem data (blksiz 4096, inosz 512, v2 dirs)  
[root@ip-172-31-30-22 ec2-user]#
```

```
[root@ip-172-31-30-22 ec2-user]#  
[root@ip-172-31-30-22 ec2-user]# mount -t xfs -o nouuid /dev/xvdb1 /mnt/old_volume  
[root@ip-172-31-30-22 ec2-user]#  
[root@ip-172-31-30-22 ec2-user]# mountpoint /mnt/old_volume  
/mnt/old_volume is a mountpoint  
[root@ip-172-31-30-22 ec2-user]#
```

Note: Using -o nouuid in the mount command due to potential conflict of current EBS meta data with newly attached EBS metadata.

Now navigate to .ssh directory of the newly attached volume.

```
[root@ip-172-31-30-22 ~]# cd /mnt/old_volume/home/ec2-user/.ssh
[root@ip-172-31-30-22 ~]# ls
authorized_keys
```

Generate a new key value pair as follows. It will generate two files <new-key> and <new-key.pub>

Save the <new-key> contents as a .pem file on your local machine as it is the private key you would use to ssh.

```
[root@ip-172-31-30-22 .ssh]#
[root@ip-172-31-30-22 .ssh]# ssh-keygen -t rsa -b 2048 -f new-key
Generating public/private rsa key pair.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in new-key
Your public key has been saved in new-key.pub
The key fingerprint is:
SHA256:zxuuufyTD/mr0X+XDcIB660looZK4s7T82s1w/BTG0 root@ip-172-31-30-22.ec2.internal
The key's randomart image is:
+---[RSA 2048]-----+
|
|   .
|  . E .
| + . o .
| + S . o .
|
| + . + . . . .
|..+..+ =. . . .
|oo * .,o*==
|o=== .+oO@=+.
+---[SHA256]-----+
[root@ip-172-31-30-22 .ssh]# ls
authorized_keys  new-key  new-key.pub
[root@ip-172-31-30-22 .ssh]#
[root@ip-172-31-30-22 .ssh]#
```

[illegible]

Replace the contents of the `authorized_keys` file with the contents of the `<new-key.pub>` file using the `nano` command.

```
[root@ip-172-31-30-22 .ssh]#  
[root@ip-172-31-30-22 .ssh]#  
[root@ip-172-31-30-22 .ssh]#  
[root@ip-172-31-30-22 .ssh]# cat new-key.pub  
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCS9vLwRbQbQ2Abf/+Luf4zbyT8FMHJdey4Rn4fJwUlaIhTh6qSE9fw0P//CRsZtTaM5aImSuFd1s3pWExfqYY+Hvw  
nSM49HLv6AyEryAgUCmc63JnH75WxxzNRShuULTYX83ie7AggTnnoYi1Yv9G+BWlQ7z4pQmxxumIpvGKQHE81fXQAVKDIZvxt0HUqkdaLAXSt1rvkXPl0IS8THP8  
al  
[root@ip-172-31-30-22 .ssh]#  
[root@ip-172-31-30-22 .ssh]# nano authorized_keys
```

```
6. ec2-54-91-5-113.compute-1.amazonaws.com X
GNU nano 5.8
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQCS9vLWrBqBQ2Abf/+LUf4zbyT8FMHJdey4Rn4fJwU1AiTh6qSE9fw0P//CRsZtTaM5a1MSuFdis3pw authorized keys
```

Remove the extra files.

```
[root@ip-172-31-30-22 .ssh]#
[root@ip-172-31-30-22 .ssh]#
[root@ip-172-31-30-22 .ssh]# ls
authorized_keys  new-key  new-key.pub
[root@ip-172-31-30-22 .ssh]#
[root@ip-172-31-30-22 .ssh]#
[root@ip-172-31-30-22 .ssh]# rm new-key
rm: remove regular file 'new-key'? y
[root@ip-172-31-30-22 .ssh]# rm new-key.pub
rm: remove regular file 'new-key.pub'? y
[root@ip-172-31-30-22 .ssh]#
[root@ip-172-31-30-22 .ssh]# ls
authorized_keys
[root@ip-172-31-30-22 .ssh]#
[root@ip-172-31-30-22 .ssh]#
```

Now detach the new volume and attach back to previous instance at the root dir.

Start the instance, and try connecting using ssh with new .pem file.

Boom! You are in!